BIMM-143: INTRODUCTION TO BIOINFORMATICS

The find-a-gene project assignment
http://thegrantlab.org/bimm143
Dr. Barry Grant

Version: 2025-04-07 (13:23:59 PDT on Mon, Apr 07)

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Overview:
The find-a-gene project is a required assignment for BIMM-143. You should prepare a written report in **PDF** format that has responses to each question labeled **[Q1] - [Q10]** below. You may wish to consult the scoring rubric at the end of this document and the example report provided online (note that the example report is from a previous quarter and the questions may differ).

The objective with this assignment is for you to demonstrate your grasp of database searching, sequence analysis, structure analysis and the R environment that we have covered in class.

Due Date:
Your responses to questions Q1-Q4 are due at 12pm on the **Monday of Week 5** (see the Assignments and Grading section of our website for details). Note that these first set of answers can be obtained very quickly (at best within 15 or 20 minutes), so if you don't succeed at first, just keep trying.

The complete assignment, including responses to all questions, is due at 12pm on the **Monday of Week 10.**

Submission instructions:
Your report formatted as a **PDF document** should be uploaded to ***GradeScope***. Please make sure to include your UCSD email and PID number on the first page.

Be sure to include your UCSD email and PID number on the first page of your report.

Submit your preliminary report with answers to Q1-Q4 as soon as you can so we can determine if you have found a novel gene. Submit this preliminary report as one document with screen shots of the results inserted appropriately.

See the demonstration report linked to on the course website for an example of format. Note again that example questions may differ. I will indicate on GradeScope my decision (1pt indicating all is good, 0pts revisions required). You should proceed with subsequent questions only after we are sure you have found a novel gene (and thus be successful in the later stages of the project).

For the final report add your results for Q5-Q10 to the preliminary report and submit the final document containing your results for all questions.

Please do not send only Q5-Q10 answers as the final report.

Questions:

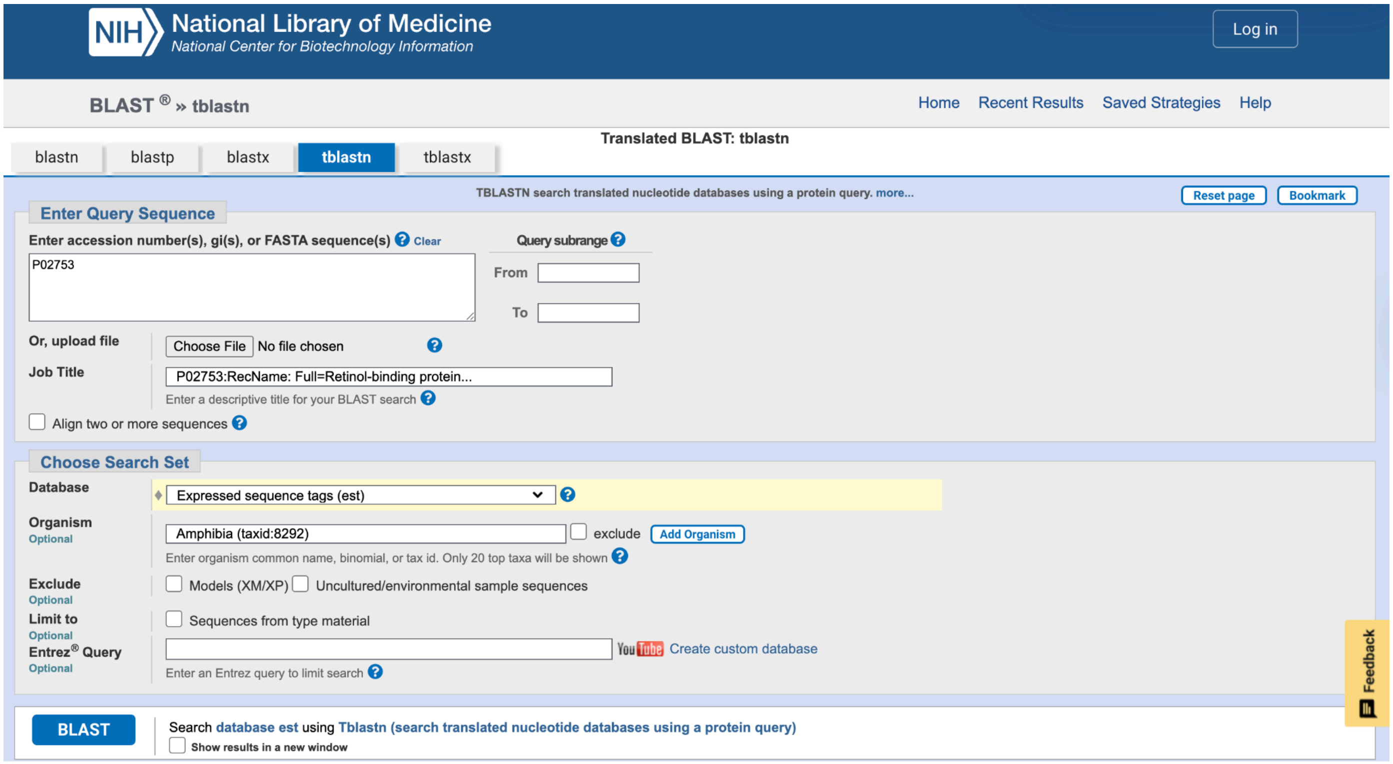
[Q1] Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Protein name: Retinol-binding protein 4 (RBP4)
Species: Homo sapiens
Accession number: P02753
Known Function: Retinol binding and retinol transmembrane transporter activity

[Q2] Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN (2.17.0+) search against amphibia ESTs
Database: Expressed Sequence Tags (est)
Organism: Amphibia (Taxid: 8292)

Also include the output of that BLAST search in your document. If appropriate, change the font to Courier size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press ⌘-shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.



On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen Match: (Result 4) Accession CN058168.1, a 785 base-pair cDNA,mRNA sequence from *Ambystoma tigrinum*. See below for alignment details. Other details include:
Query coverage = 90%
E-value = 4e-93
% identity = 70%
Accession length = 785

Your search is limited to records **include: Amphibia (taxid:8292)**

Job Title	**P02753:RecName: Full=Retinol-binding protein...**
RID	RUSP8CKS014 *Search expires on 02-01 09:02 am* Download All
Program	TBLASTN Citation
Database	est See details
Query ID	P02753.3
Description	RecName: Full=Retinol-binding protein 4; AltName: Full=Pl ...
Molecule type	amino acid
Query Length	201
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☑	Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	2e-93	69.61%	731	CN064263.1
☑	Salamander_10_I03.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-93	70.00%	735	CN052017.1
☑	Salamander_20_N22.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-93	70.00%	760	CN055612.1
☑	Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	4e-93	70.00%	785	CN058168.1
☑	Ag2_p35_J10_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	2e-92	70.00%	913	CN062105.1
☑	Ag2_p36_A19_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	3e-92	70.00%	958	CN062231.1
☑	v3_p1_a3_t3 V3 Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	272	272	88%	5e-92	70.62%	697	CN049128.1
☑	Ag2_p38_K1_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	6e-92	70.00%	1012	CN062855.1
☑	Ag2_p38_2_D23_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma ti...	276	276	90%	1e-91	70.00%	1094	CN062829.1

Alignment Details:

Download GenBank Graphics ▼ Next ▲ Previous ◀ Descriptions

Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence

Sequence ID: CN058168.1 **Length:** 785 **Number of Matches:** 1

Range 1: 104 to 643 GenBank Graphics ▼ Next Match ▲ Previou

Score	Expect	Method	Identities	Positives	Gaps	Frame
276 bits(706)	4e-93	Compositional matrix adjust.	126/180(70%)	153/180(85%)	0/180(0%)	+2

```
Query  14   GSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFL
             S  AERDC VS+F V +NFD+ R++GTWYAMAKKDPEGLFL
Sbjct  104  ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFL

Query  74   TAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWGVASFL
            TAKGRV +   + +CADMVG FTDT+DPAKF M+Y G+ ++L
Sbjct  284  TAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAYL

Query  134  VQYSCRLLNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQE
            + YSCR LN DGTCADSYSFVFSRDPNGL PEAQ+IVR+RQ+
Sbjct  464  LHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQK
```

Alignment summary:
Score: 276 bits (706)
Expect: 4e-93
Method: Compositional matrix adjust.
Identities: 126/180 (70%)
Positives: 153/180 (85%)
Gaps: 0/180 (0%)
Frame: +2

In general, [Q2] is the most difficult for students because it requires you to have a “feel” for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not “novel”), a near match (something that might be “novel”, depending on the results of [Q4]), and a non-homologous result. If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

[Q3] Gather information about this “novel” protein. At a minimum, show me the protein sequence of the “novel” protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don’t forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don’t have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Chosen sequence (from EMBOSS TRANSEQ)-

```
>104-643_1 Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLATAKGRVVIFGGFQL
CADMVGIFTDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYSFVFSRDPNG
LTPEAQRIVRKRQKELCLDREYRMVVHNGYCE
```

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as S. cerevisiae, human or mouse, because those genomes have already been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: Ambystoma novel protein
Species: *Ambystoma tigrinum*
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Amphibia; Batrachia; Caudata; Salamandroidea; Ambystomatidae; Ambystoma.

[Q4] Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, “novel” is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

- If there is a match with 100% amino acid identity to a protein in the database, from the same species, then your protein is NOT novel (even if the match is to a protein with a name such as “unknown”). Someone has already found and annotated this sequence, and assigned it an accession number.
- If the top match reported has less than 100% identity, then it is likely that your protein is novel, and you have succeeded.
- If there is a match with 100% identity, but to a different species than the one you started with, then you have likely succeeded in finding a novel gene.

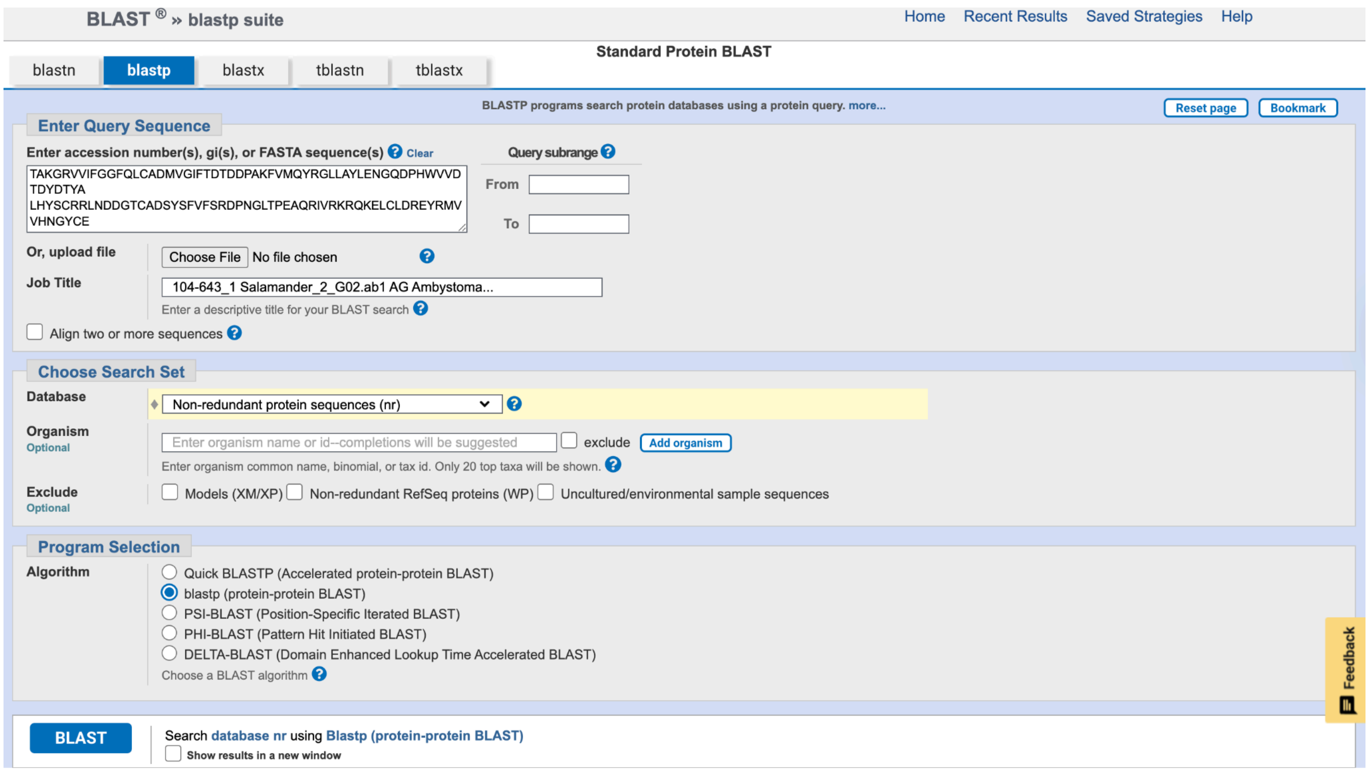
• If there are no database matches to the original query from [Q1], this indicates that you have partially succeeded: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:

A BLASTP search against the NR database (see setup in first screenshot below) yielded a top hit result is to a protein from *Ambystoma mexicanum* (Axolotl). Even though it has a 100% identity, it is from a different species than the one we started with (*Ambystoma mexicanum* vs *Ambystoma tigrinum*)

Other details include 100% coverage and an E-value of 7e-132.

See additional screenshots below for top hits and selected alignment details.



‹ Edit Search | Save Search | Search Summary ⌄ | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Job Title	104-643_1 Salamander_2_G02.ab1 AG Ambystoma...
RID	RV7J166R014 *Search expires on 02-01 13:16 pm* Download All ⌄
Program	BLASTP Citation ⌄
Database	nr See details ⌄
Query ID	lcl|Query_4859546
Description	104-643_1 Salamander_2_G02.ab1 AG Ambystoma tigrinu ...
Molecule type	amino acid
Query Length	180
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Compare these results against the new Clustered nr database BLAST

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download ⌄ Select columns ⌄ Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
retinol-binding protein 4 [Ambystoma mexicanum]	Ambystoma mexicanum	378	378	100%	7e-132	100.00%	200	XP_069467453.1
retinol-binding protein 4 [Ascaphus truei]	Ascaphus truei	298	298	99%	3e-100	77.09%	195	XP_075468936.1
retinol-binding protein 4 [Pleurodeles waltl]	Pleurodeles waltl	295	295	99%	6e-99	75.42%	199	XP_069095936.1
retinol-binding protein 4 [Lissotriton helveticus]	Lissotriton helveticus	294	294	99%	2e-98	76.54%	200	XP_078499520.1
retinol-binding protein 4 [Geotrypetes seraphini]	Geotrypetes seraphini	291	291	99%	2e-97	73.74%	199	XP_033799662.1
retinol-binding protein 4 [Rhinatrema bivittatum]	Rhinatrema bivittatum	290	290	99%	8e-97	73.18%	201	XP_029465754.1

See alignment details here:

Download ⌄ GenPept Graphics ▼ Next ▲ Previous ◀ Descriptions

retinol-binding protein 4 [Ambystoma mexicanum]

Sequence ID: XP_069467453.1 **Length: 200** **Number of Matches: 1**

See 1 more title(s) ⌄ See all Identical Proteins(IPG)

Range 1: 16 to 195 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
378 bits(971)	7e-132	Compositional matrix adjust.	180/180(100%)	180/180(100%)	0/180(0%)

```
Query  1    ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLA  60
            ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLA
Sbjct  16   ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLA  75

Query  61   TAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYA  120
            TAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYA
Sbjct  76   TAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYA  135

Query  121  LHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVVHNGYCE  180
            LHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVVHNGYCE
Sbjct  136  LHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVVHNGYCE  195
```

Related Information

Genome Data Viewer - aligned genomic context

Identical Proteins - Identical proteins to XP_069467453.1

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width. Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Relabeled sequences for alignment:

```
>Homo_sapiens_RBP4
MKWVWALLLLAALGSGRAERDCRVSSFRVKENFDKARFSGTWYAMAKKDPEGLFLQDNIV
AEFSVDETGQMSATAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWGVASFLQKGND
DHWIVDTDYDTYAVQYSCRLLNLDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLA
RQYRLIVHNGYCDGRSERNLL

>Ambystoma_novel_RBP4
ASCLAERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLATAKGRVVIFGGFQLCADMVGIFTDTD
DPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIVRKRQKELCLDREYRM
VVHNGYCE

>Dendrobates_RBP4
MDLKVFGLLIALCYFGSTEAFRFYERDCTVSNFRVMPNFDKTRYAGTWYAVAKKDPEGLFLIDNIVANFN
VDEDGKMTATAKGRVVIFQTMELCADMVGTFQETDDPAKFKMKYHGVLSFLEKGLDDHWVVDTDYDNYAV
HYSCREVDEDGTCLDSYSFIFSRDENGLTPEIQRIVRRRQEELCLDRKYRVVKHNGYCTQT

>Pangshura_RBP4
MAQAGRFLAWLLLVALGLLGGGCQAERDCRVSNFRVQENFDKARYTGTWYAMAKKDPEGLFLQDNVVAQF
TIDENGQMSATARGRVRLFNEWDVCADMIGSFTDTEDPAKFKMKYWGVASFLQKGNDDHWVIDTDYDTYA
LHYSCRQLNDDGTCADSYSFVFSRDPKGLPPEAQRIVRQRQVDLCLDRKYRVIVHNGFCP

>Cygnus_atratus_RBP4
MAYTRKALPCLLLLALTFLGSSMAERDCRVSSFKVKENFDKNRYSGTWYAMAKKDPEGLFLQDNVVAQFT
VDENGQMSATAKGRVRLFNNWDVCADMIGSFTDTEDPAKFKMKYWGVASFLQKGNDDHWVVDTDYDTYAL
HYSCRQLNEDGTCADSYSFVFSRDPKGLPPEAQKIVRQRQIDLCLDRKYRVIVHNGFCS
```

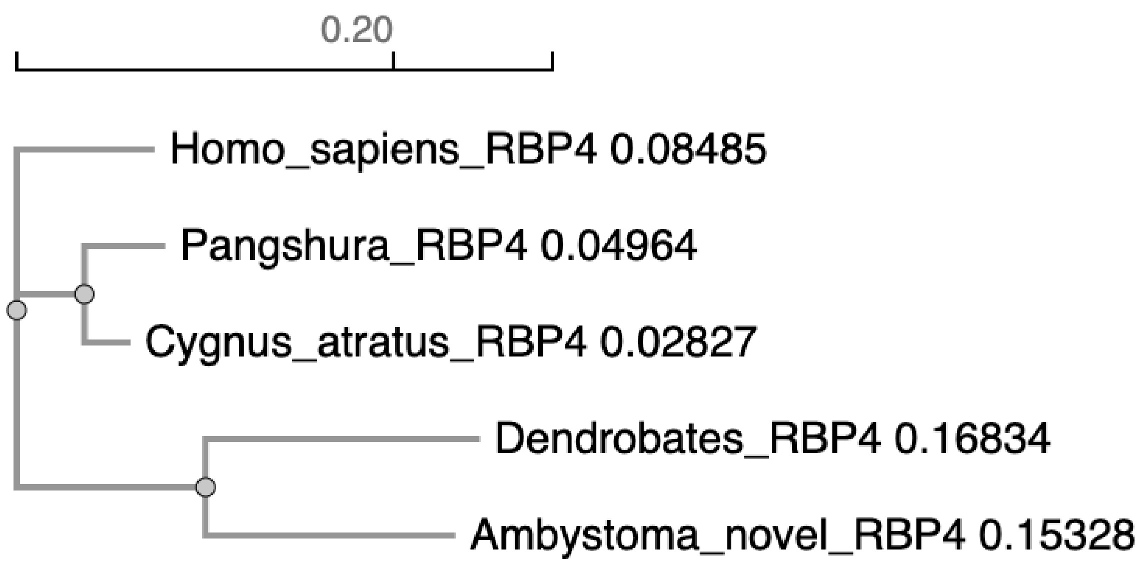
Alignment:

Obtained using MUSCLE (version 3.8) at EBI

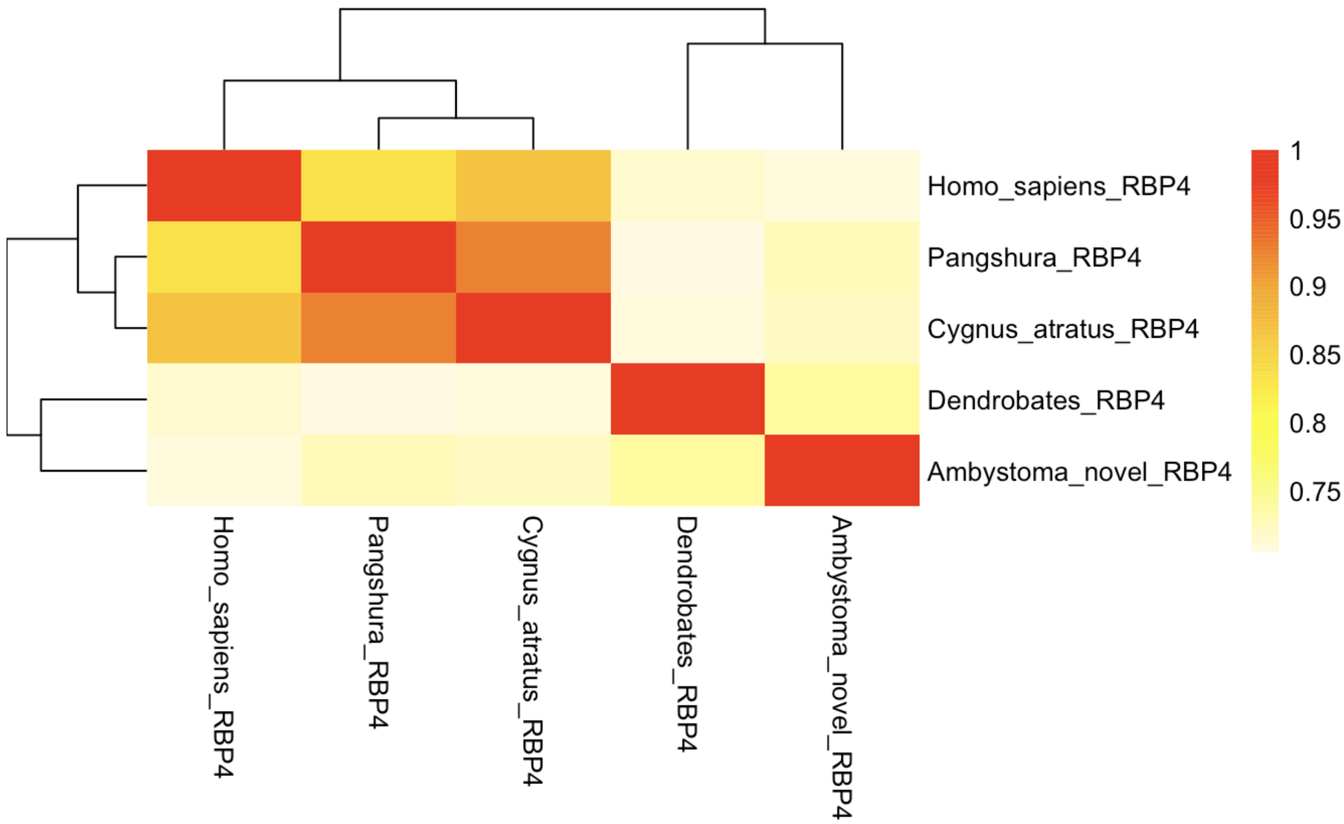
```
CLUSTAL multiple sequence alignment by MUSCLE (3.8)


Homo_sapiens_RBP4         AERDCRVSSFRVKENFDKARFSGTWYAMAKKDP
Pangshura_RBP4            AERDCRVSNFRVQENFDKARYTGTWYAMAKKDP
Cygnus_atratus_RBP4       AERDCRVSSFKVKENFDKNRYSGTWYAMAKKDP
Dendrobates_RBP4          YERDCTVSNFRVMPNFDKTRYAGTWYAVAKKDP
Ambystoma_novel_RBP4      AERDCTVSNFPVMQNFDRYRYTGTWYAMAKKDP
                           **** **.* *  ***. *::*****:*****

Homo_sapiens_RBP4         EGLFLQDNIVAEFSVDETGQMSATAKGRVRLLNNWDVCADMVGTFTDTEDPAKFKMKYWG
Pangshura_RBP4            EGLFLQDNVVAQFTIDENGQMSATARGRVRLFNEWDVCADMIGSFTDTEDPAKFKMKYWG
Cygnus_atratus_RBP4       EGLFLQDNVVAQFTVDENGQMSATAKGRVRLFNNWDVCADMIGSFTDTEDPAKFKMKYWG
Dendrobates_RBP4          EGLFLIDNIVANFNVDEDGKMTATAKGRVVIFQTMELCADMVGTFQETDDPAKFKMKYHG
Ambystoma_novel_RBP4      EGLFLLDNIVAKFTVDDYGRMLATAKGRVVIFGGFQLCADMVGIFTDTDDPAKFVMQYRG
                          ***** **:**:*.:*: *.* ***.*** ::   ::****:* * :*:***** *:*.*

Homo_sapiens_RBP4         VASFLQKGNDDHWIVDTDYDTYAVQYSCRLLNLDGTCADSYSFVFSRDPNGLPPEAQKIV
Pangshura_RBP4            VASFLQKGNDDHWVIDTDYDTYALHYSCRQLNDDGTCADSYSFVFSRDPKGLPPEAQRIV
Cygnus_atratus_RBP4       VASFLQKGNDDHWVVDTDYDTYALHYSCRQLNEDGTCADSYSFVFSRDPKGLPPEAQKIV
Dendrobates_RBP4          VLSFLEKGLDDHWVVDTDYDNYAVHYSCREVDEDGTCLDSYSFIFSRDENGLTPEIQRIV
Ambystoma_novel_RBP4      LLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYSFVFSRDPNGLTPEAQRIV
                          : ::*::* * **::*****.**::**** :: **** *****:**** :**.** *.**

Homo_sapiens_RBP4         RQRQEELCLARQYRLIVHNGYCD
Pangshura_RBP4            RQRQVDLCLDRKYRVIVHNGFCP
Cygnus_atratus_RBP4       RQRQIDLCLDRKYRVIVHNGFCS
Dendrobates_RBP4          RRRQEELCLDRKYRVVKHNGYCT
Ambystoma_novel_RBP4      RKRQKELCLDREYRMVVHNGYCE
                          *.** :*** *:**:: ***:*
```

[Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use "simple phylogeny" online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

Neighbor joining (distance-based) phylogenetic tree created using "simple phylogeny" online from the EBI.



[Q7] Generate a sequence identity based heatmap of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and “Save as” FASTA format for example). Read this FASTA format alignment into R with the help of functions in the Bio3D package. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.



[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences. List the top 3 unique hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

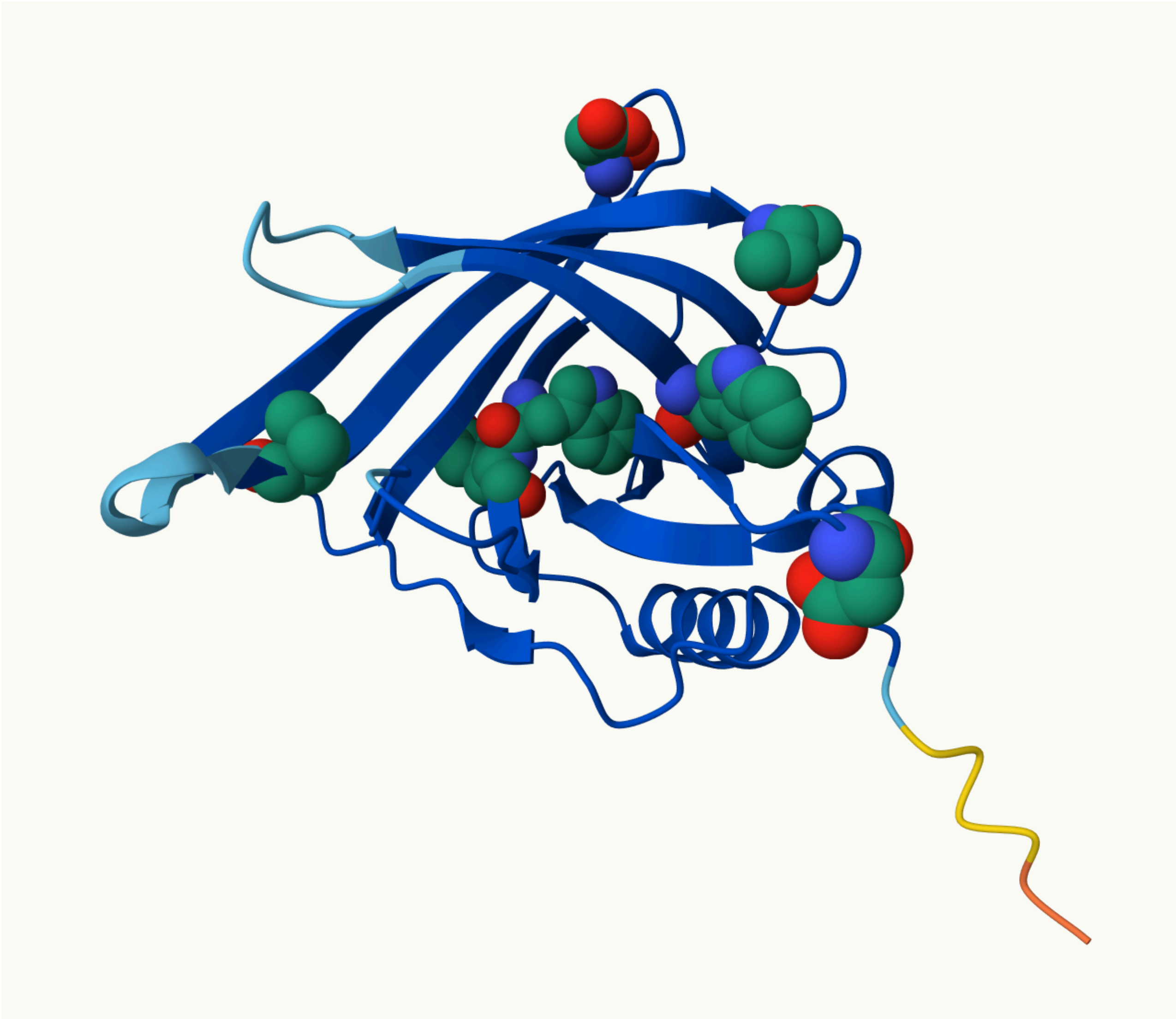
HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function consensus(). The Bio3D functions blast.pdb(), plot.blast() and pdb.annotate() are likely to be of most relevance for completing this task. Note that the results of blast.pdb() contain the hits PDB identifier (or pdb.id) as well as Evalue and identity. The results of pdb.annotate() contain the other annotation terms noted above. Note that if your consensus sequence has lots of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by calculating the row-wise maximum from your sequence identity matrix.

ID	**Technique**	**Resolution**	**Source**	**E-value**	**Identity**
1IIU	X-RAY DIFFRACTION	2.5	Gallus gallus	1.68e-130	99.4
4O9S	X-RAY DIFFRACTION	2.3	Homo sapiens	3.71e-118	87.4
6QBA	X-RAY DIFFRACTION	1.8	Saccharolobus solfataricus	9.33e-118	87.4

[Q9] Using AlphaFold notebook generate a structural model using the default parameters for your novel protein sequence.

> Note that this can take some time depending upon your sequence length. If your model is taking many hours to generate or your input sequence yields a "too many amino acids" (i.e. length) error you can focus on a single domain from your sequence - identify region by searching for PFAM domain matches.

Once complete save the resulting PDB format file for your records. Finally, generate a molecular figure of your generated PDB structure using the Mol* viewer online (or VMD/PyMol/Chimera if you prefer). To complete your analysis you should highlight conserved residues that are likely to be functional as spacefill and the protein as cartoon colored by local alpha fold pLDDT quality score. You can determine conserved residues from the alignment generated by the AlphaFold server and use a conservation cutoff appropriate for the diversity of your protein alignment (e.g. between 60% and 99% conserved). Note that pLDDT score is contained in the B-factor column of your PDB downloaded file. Please use a white or transparent background for your figure (i.e. not the default black in PyMol/VMD/Chimera etc.).



Molecular figure created using Molstar. Cartoon is colored by AlphaFold pLDDT quality score, and spacefill residues indicate conserved residues that are likely functional (100% identity i.e fully conserved across the 5-sequence MSA).

[Q10] (i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (https://cfold.bme.uic.edu/castpfold/). Provide an image or screen-shot of your largest predicted pockets “negative volume” and provide it’s area and volume.

(ii) Perform a “Target” search of ChEMBEL (https://www.ebi.ac.uk/chembl/) with your novel sequence. Are there any Target Associated Assays and ligand efficiency data reported that may be useful starting points for exploring potential inhibition of your novel protein? If there are no assays listed here simply list “non available as of [date]”.

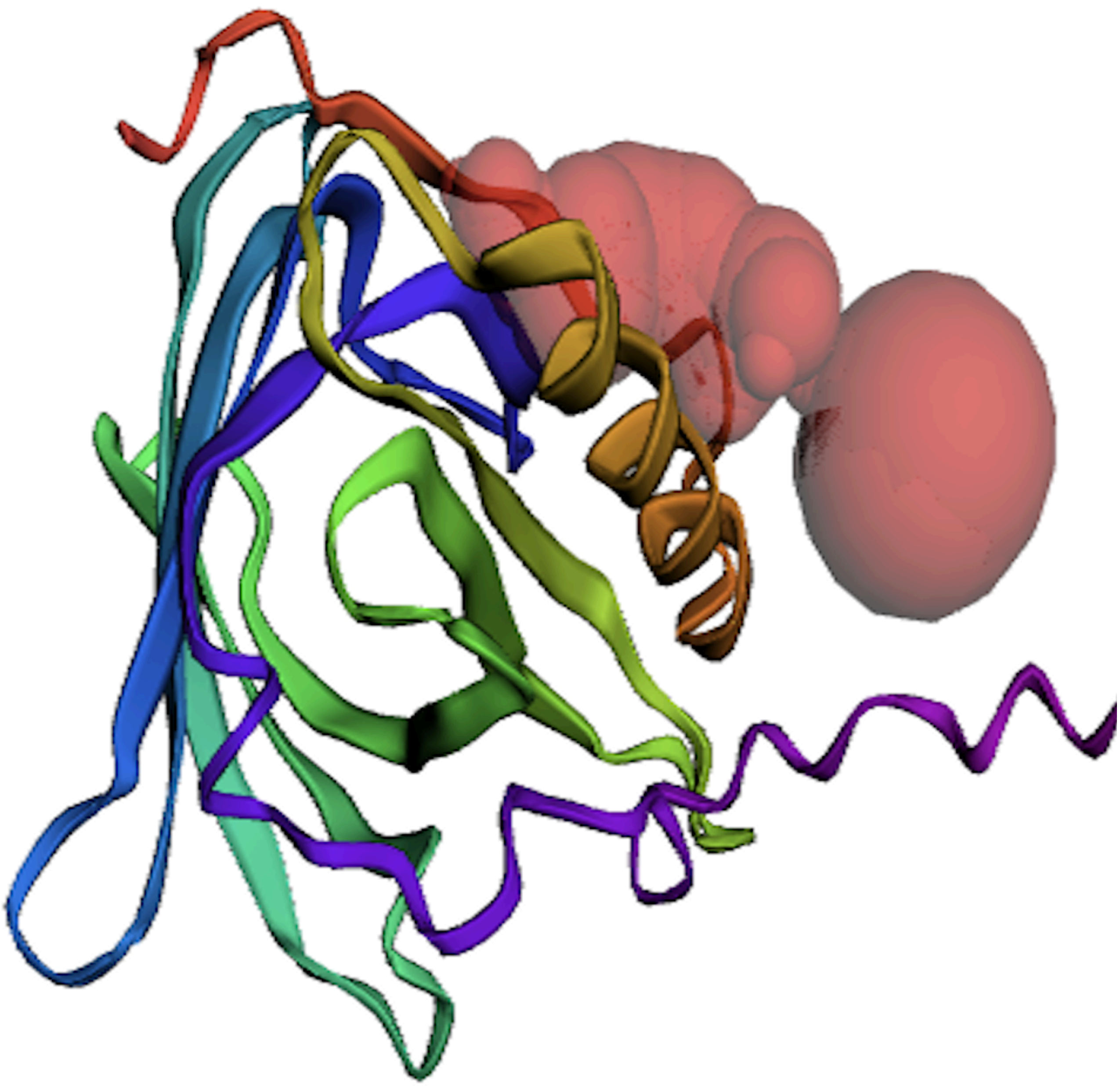
(iii) Briefly discuss (100 words max) the druggability of your novel protein based on: - Presence of well-defined pockets (output of tools like CASTpFold), - Existence of known inhibitors for related proteins (your search of ChEMBEL), - Conservation of binding sites across homologs (your conservation analysis in Q10), - Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

i) Pocket: Pocket ID 1

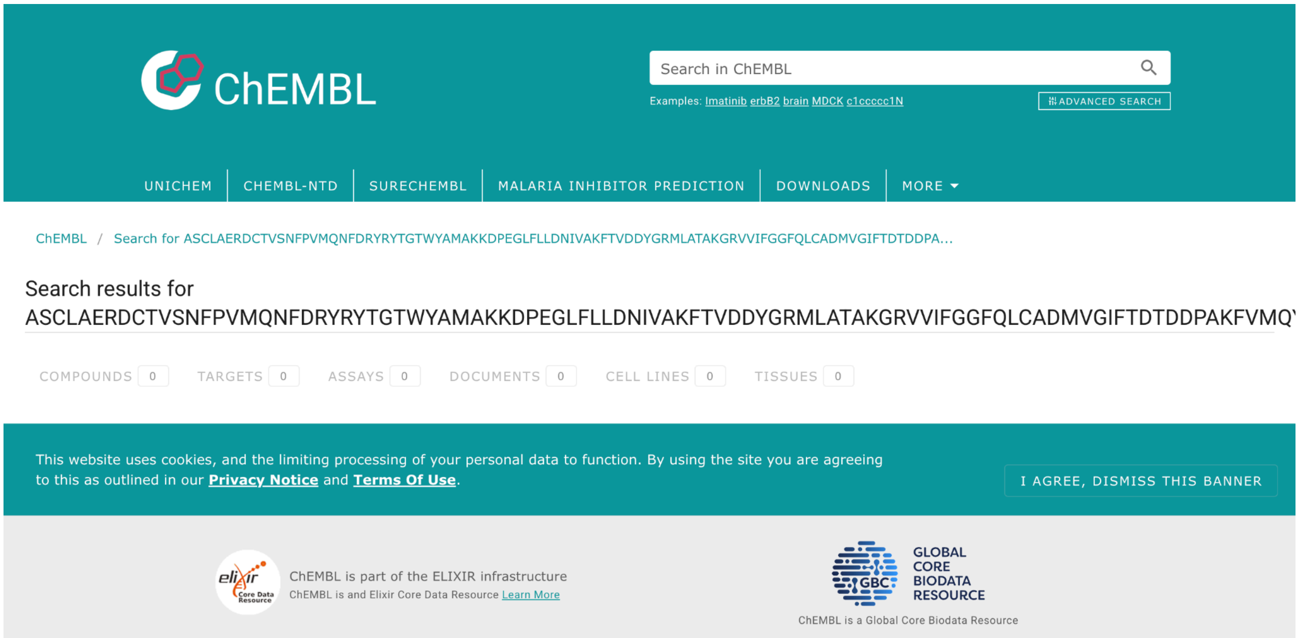
Area: 232.121 $\dot{A}^2$

Volume: 225.468 $\dot{A}^3$

CASTpFold prediction of the largest pocket (negative volume shown in red in the figure).



ii) A target search using the novel protein sequence returned no assays/targets in ChEMBL. No Target Associated Assays or ligand efficiency data were available as of Mar 8, 2026.



iii) CASTpFold identified a well-defined pocket (area: 232.121 $Å^2$, volume: 225.468 $Å^3$), consistent with a potential ligand binding cavity. The ChEMBL results did not yield Target Associated Assays/efficiency data; therefore, there are currently no inhibitors specific to this novel sequence available to leverage yet. However, the pocket lining residues (~148-172 and nearby) are moderately-to-highly conserved across homologs in MSA, suggesting a functionally important site. If this novel protein is RBP4-like, targeting this pocket could modulate retinoid transport/receptor-mediated uptake; literature links RBP4/retinoid homestasis to ocular disease and retinal degeneration, including Stargardt and age-related macular degeneration.